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The effects of video display terminal height on the operator: a comparison of the 15° and 40° recommendations

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Prolonged use of video display terminals (VDTs) has been shown to be a risk factor for musculoskeletal and visual discomfort. A standard workplace design recommendation is to position the centre of the VDT 15° below horizontal eye level. Recently a viewing angle of 40° below horizontal has been suggested based on studies that have indicated that this is the preferred viewing angle for visually intensive tasks. The goal of this study was to investigate the effects of these two VDT positions on muscular activity, muscular fatigue, head/neck posture, visual acuity, operator performance (productivity and quality), heart rate and operator subjective assessment. The experimental task consisted of reading text from a computer screen and answering reading comprehension questions using a mouse and a keyboard. Each experimental session lasted 2 h. The 40° VDT position showed significantly greater head tilt angles and higher muscle activity levels for six of the 10 neck, shoulder and back muscles sampled. No significant differences in visual acuity, operator performance or heart rate were detected as a result of monitor location. Seven of the 12 subjects preferred the 15° monitor position. © **1998 Elsevier Science Ltd. All rights reserved.**

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Introduction

By the end of 1993, 74 million computers were being used in the United States, making up 43% of the world-wide computer market (Juliussen and Petska-Juliussen, 1994). The widespread use of computers and video display terminals (VDTs) has resulted in an increasing concern for the physical well-being of the user, specifically the musculoskeletal and visual systems. Accordingly, considerable research on positioning the VDT to minimize discomfort and fatigue has been undertaken (Arndt, 1983; Jaschinski-Kruza, 1988, 1991; Krueger, 1984; Pickett and Lees, 1991; Sauter *et al*, 1991; Villanueva *et al*, 1996) Despite this effort, 10 million cases of VDT-related eyestrain and physical discomfort are reported to the American Optometric Association each year (Leavitt, 1995). Though many of these complaints arise from workstations that are incorrectly adjusted for the individual, Arndt (1983) found that complaints may still arise from individuals working with properly adjusted workstations.

Leavitt (1995) reported that eyestrain symptoms may occur in up to 75% of all VDT users, while Pickett and Lees (1991) found the percentage reporting eyestrain to be as high as 85%. Tyrrell and Leibowitz (1990) described eyestrain as a 'vague discomfort which may be localized in the head or eyes', and included eyestrain in a larger category termed visual fatigue. Visual fatigue can be described as any subjective visual symptom or distress resulting from the use of ones eyes (Tyrrell and Leibowitz, 1990). Visual fatigue during VDT use is often caused by prolonged visual activity performed during near-work. Near-work causes the eyes to converge and the lens to accommodate in order to focus, forcing the extraocular and ciliary muscles of the eye to contract and maintain contraction. According to Jaschinski-Kruza (1988, 1991), this extraocular and ciliary muscle work leads to eye strain and visual fatigue. Thus, as long as VDT work causes the eyes to converge, complaints of eye fatigue can be expected to continue.

Musculoskeletal discomfort of VDT users most commonly includes pain in the back, neck, and shoulders (Arndt, 1983). Sauter *et al* (1991) reported that musculoskeletal discomfort was 'commonplace' during VDT work, finding 33% of 539 data entry VDT users reported

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discomfort in the lower back area, 27% reported discomfort in the neck and buttocks, and 15% reported pain in the right shoulder. Pickett and Lees (1991) found even higher complaint rates among 78 female VDT workers while finding no evidence that the presence of adjustable workstations offered protection against physical complaints.

VDT users often perform prolonged seated tasks that can cause static loading of the muscles of the neck and upper back. Awkward, static postures tend to reduce blood circulation to the muscles, decreasing the flow of nutrients and removal of wastes, which can lead to muscle fatigue and pain (Arndt, 1983). Static postures can also lead to elevated heart rate (Sanders and McCormick, 1993) and cause chronic problems in joints and tendons (Grandjean, 1980). While VDT users are often advised to take frequent breaks that include movement and stretching, most operators do not strictly follow these guidelines, but instead, simply alter their seated posture (either consciously or subconsciously) by shifting around in their chair. The postures resulting from these attempts to alleviate static loading are often awkward and have the potential to cause as much discomfort as the original static posture (Watson and Trott, 1993; Kilbom *et al* 1986; Kumar, 1994). One basic VDT workplace design principle is to position the monitor such that the posture that the operator assumes will not require significant static muscle forces that will result in a reduction of blood flow.

An often encountered recommendation for VDT placement is to place the topmost line of the viewing monitor at or slightly below eye level (OSHA, 1991; US Department of Labor, 1995). This positioning strategy places the centre of the monitor approximately 15° below horizontal eye level and has been recommended because it allows the operator to assume a comfortable and natural posture, as the normal line of sight is actually 15° below horizontal (Sanders and McCormick, 1993; Eastman Kodak Human Factors Section, 1983).

Eye gaze angle and visual convergence have been shown to be two important areas of research in VDT workplace design. Eye gaze angle has been reported to be dependent on head inclination angle, as eye gaze angle can be adjusted by varying the head angle and/or the inclination of the eye (Heuer and Owens, 1989). Previous research studies have found that eyes tend to diverge as they are elevated and converge as they are lowered (Hering, 1977; Hemholtz, 1962; Heuer and Owens, 1989). The results of these previous studies showed that natural near-distance convergence occurs when the gaze angle is below horizontal while divergence is found with elevated gaze angles. Both of the monitor placement angles tested within this current study allow for natural near-distance convergence by positioning the centre of the monitor at or below the natural line of sight.

Due to the large numbers of visual, postural and musculoskeletal complaints still surrounding VDT workstations, new ideas concerning workstation design are beginning to surface. One workstation modification is to lower the VDT to a position beyond 15° so that the centre of the monitor is actually below the work surface. It has been suggested that 40° is the preferred viewing angle for seated work (Eastman Kodak Human Factors Section, 1983) and for visual tasks such as VDT use (Kroemer and Hill, 1986). The lower position may aid vision by improving eye moisture. Sotoyama *et al* (1996) reported that downward gaze angles decrease the surface area of the eye exposed to the atmosphere, thus reducing dry eye problems that arise during VDT use. Other research reports blink rate as a reliable indicator of dry eyes and visual discomfort (Villanueva *et al*, 1996). Beyond visual recommendations, Ankrum and Nemeth (1995) postulated that the user is able to adopt a wider range of head tilt angles while viewing the monitor at 40°, thereby reducing static loading of the muscles of the neck, shoulder and back. Conversely, as the monitor is placed below the work surface, deviated neck postures and resulting elevated muscle activity are a concern. Fatigue may also become a problem if head tilt angles associated with use of lower monitors remain moderately static over the course of VDT use.

Although some theoretical exploration supports the idea of lowering the monitor, limited research has actually investigated the effects of a 40° position. The goals of the present study are to examine the biomechanical, performance, physiological and visual effects on the user of a 40° and a 15° monitor position.

Methods

Subjects

Subjects were recruited from North Carolina State University and surrounding populations. Only individuals who had no history of a significant back, neck or shoulder injury were considered as potential subjects. An injury was considered significant when it required lost time from work, school or regular activities. Five male and seven female subjects participated in the study. The average age, height and weight of the subjects were 26.1 (SD = 4.4) yr, 174.1 (SD = 8.7) cm, 77.9 (SD = 18.2) kg, respectively. Seven of the subjects had uncorrected vision and five of the subjects required single-vision contacts. The experiment was performed in two sessions with a total duration of approximately 5 and $\frac{1}{2}$ h.

Apparatus

The experiment was conducted at a fully adjustable VDT workstation, see *Figure 1*. The seat height, table height, monitor height and monitor angle were all positioned according to each subject's anthropometric measurements. Seat height was calculated by adding two centimeters to the subject's popliteal height to allow for seat compression (Bendix, 1987). However, once the subject

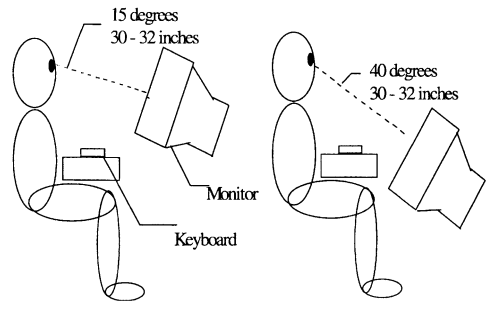


Figure 1 Workstation configuration

was seated, slight seat height adjustments were made to ensure the subject's comfort and to maintain a 90° knee flexion angle as suggested by Chaffin and Andersson (1991). Keyboard support surface was adjusted for each subject so that the elbow angle was approximately 90°. (Chaffin and Andersson, 1991; Sanders and McCormick, 1993). The monitor height and tilt angle differed for each subject depending on monitor placement condition, 15 or 40°. Monitor height, the distance from the floor to the centre of the screen, was based on the seated eye height of the subject. A 30–32 in viewing distance for both conditions was chosen to reduce eye convergence and accommodation in an effort to alleviate visual fatigue (Krueger, 1984; Jaschinski-Kruza, 1991). The tilt angle of the monitor was set such that the line of sight was perpendicular to the screen surface.

The workstation was located in a room that controlled for all types of glare. The windows were covered by closed blinds and the room was illuminated with fluorescent lamps that were not located directly over the workstation. No visible direct or reflected glare was found on the screen at any time. In both conditions, ambient illumination was approximately 475 lx and was measured at table height, which is within the recommended ANSI (1988) standard range of 200–500 lx for VDT working areas.

Each condition for each subject was videotaped on two orthogonally placed video cameras. A camera placed 3 m behind the subject recorded all movements in the coronal plane, while the camera positioned 2 m to the right of the subject recorded all movements in the sagittal plane. A background grid was placed in front of and to the left of the subject to provide permanent landmarks during analysis.

Myoelectric activity was collected using 21 bipolar surface electrodes. The Ag–Ag/Cl electrodes (model E22x, InVivo Metric, Healdsburg, CA) had an inner diameter of 4 mm. The specific collection sites for the 10 muscles were determined using submaximal exertions and palpation. The collection sites were located over the belly of the muscle or on a site where the muscle was most superficial. The 10 electrode pairs, with a 2 cm inter-electrode distance, were positioned in line with the muscle fibres. A ground electrode was placed on the acromion process of the subject's right shoulder.

The pre-amplified signal was carried via shielded cables to the electromyographic (EMG) system (Data Design, Inc., Columbus, Ohio). The EMG signal was filtered so that signals above 2000 Hz were eliminated. This filtered signal was then rectified and smoothed using an integration constant of 50 ms. The integrated signal was converted from analog-to-digital form using a converter running on an IBM compatible 486DX personal computer. The raw signal, bypassing the rectification and smoothing operations, was also collected.

For maximum voluntary contractions and sub-maximal experimental test contractions, a Kin Com isokinetic dynamometer was used (Chattecx Corporation, Chattanooga, TN). The dynamometer provided the ability to position the joint of interest in the appropriate postures in order to measure the maximum forces produced. During the submaximal exertions the subjects utilized the visual feedback capabilities of the apparatus to control their force output. These techniques will be described in greater detail in the Procedure section.

Experimental task

The task consisted of reading text of 12 pt. Arial font, black letters on white background, presented on a 13 in VDT. The character height provided the preferred range of 20–22 min of arc, optimizing both readability and legibility (ANSI/HFS 100-1988). The reading text, *Wuthering Heights* by Emily Bronte, was selected for its moderate reading level (Flesch Reading Ease Score of 54.7) and interest factor for the average college student. Speed of text presentation was controlled by the subject with mouse or keyboard and at the end of each chapter two questions were presented that required short answers or multiple choice selection. A program recorded the time spent reading each chapter and answering the corresponding questions for measures of performance. The task was designed with minimal typing and mousing to reduce muscle and postural contributions arising from task related movement.

Experimental design

Independent variable. Two monitor placements were used for this study, 15° and 40° below the horizontal line of sight. Two 2 h sessions, each utilizing one monitor height, were scheduled for each subject using a counterbalanced design in which half of the subjects worked at the 15° monitor position first, while the other half began with the 40° position.

Dependent variables. Dependent variables are listed below:

(1) *Muscle activity*. Integrated EMG data were used to quantify the activity of the sampled muscles. In order to compare activity within and across subjects, these integrated EMG data were normalized with respect to maximal voluntary contraction EMG levels which were collected at the beginning of the experiment. The 10 muscles sampled were: the right trapezius (RTRAP), left trapezius (LTRAP), right cervical erector spinae (RCERV), left cervical erector spinae (LCERV), right thoracic erector spinae (RTHOR), left thoracic erector spinae (LTHOR), right sternocleidomastoid (RSTERNO), left sternocleidomastoid (LSTERNO), right levator scapulae (RLEV), left levator scapulae (LLEV). A ground electrode was placed on the acromion process of the right scapula. The sampling locations are illustrated in *Figure 2*. Muscle activity was sampled at 5 min increments throughout the task.

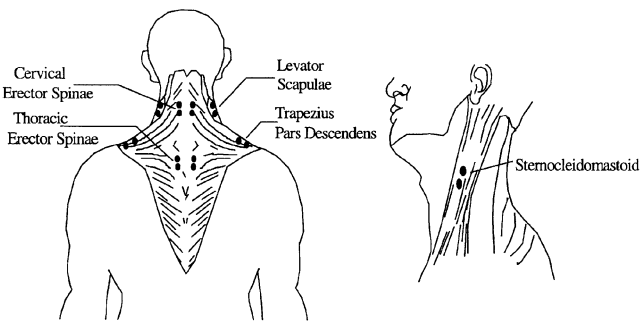


Figure 2 Electrode placement locations

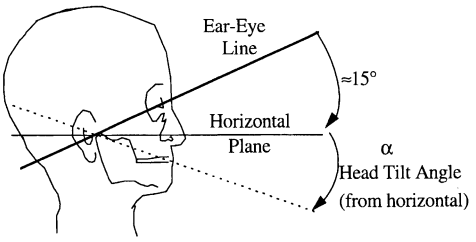


Figure 3 Head tilt angle diagram

(2) *Muscle fatigue* One second test contractions, equaling 20% of the subject's maximum voluntary contraction, were performed every 15 min throughout the 2 h task. A Fast Fourier Transform (FFT) algorithm was employed to analyse the raw EMG data and calculate median frequencies of the raw EMG signal as a function of time. Any downward shift in median frequencies was explored as a possible indicator of fatigue.
(3) *Posture.* Fluorescent dots were placed at seven locations on the subject: centre of back of head (occiput), right shoulder and left shoulders (acromion processes), base of neck (C7), upper back (T6), canthus of right eye, and tragus of right ear. Head tilt and number of postural shifts (anterior–posterior and lateral) were evaluated by use of video analysis at three time points during the task (5, 65, 110 min). Head tilt was calculated by measuring the angle between the horizontal and the ear–eye line and then subtracting 15°, which is the natural postural deviation (Jampel and Shi, 1992) (*Figure 3*).
(4) *Blink rate.* The number of blinks per session were recorded while viewing the video of the subject's session. These were sampled in 5 min intervals, three different times throughout the experiment at 5–10, 65–70 and 110–115 min.
(5) *Visual acuity.* Subjects's visual acuity was tested upon completion of each session using a Titmus Vision Tester (Model OV-7M) and a Landholt rings test.
(6) *Performance.* Performance was assessed along two dimensions: quality and productivity. The percentage of correct answers was considered a measure of quality of work, while productivity considered the time involved with reading each chapter and answering the respective questions.
(7) *Heart rate.* Heart rate was monitored using a heart rate monitor (Polar) and was sampled at 15 min intervals throughout the experiment.
(8) *Subjective assessment/discomfort questionnaire.* At the conclusion of the experiment the subjects were asked to complete a questionnaire pertaining to workstation configuration preferences and discomforts encountered during the experiment. The questionnaire consisted of four questions pertaining to workstation design preferences followed by a table of body parts and a rating scale [1 (barely noticeable) to 5 (painful)] to aid subjects in accurately describing any discomfort felt during the experiment. The subject was also asked to discriminate between new, aggravated or existing pain or discomfort.

Procedure

The subjects participated in two experimental sessions which included preparatory information, measurement of anthropometry (Session #1 only), electrode placement, maximal voluntary contractions, the 2 h experimental reading task, and concluding questions.

Session #1. The subject was informed of the requirements of the experiment and asked to carefully read and sign a consent form. Basic anthropometric measures were taken. Using the subject's anthropometric measurements, workstation dimensions were calculated. The subject's neck and upper back were then prepared for electrode placement, using established guidelines (Marras, 1990), and the electrodes were affixed to the subject's skin.

A battery of seven different maximal isometric exertions were performed by each subject. The subjects were seated in the task chair and instructed to keep both feet flat on the floor while concentrating on using only the neck and upper back muscles to exert maximum force. Verbal encouragement was given to the subjects for motivation. The first exertion required the subject to extend the neck and head against the resistant arm of the Kin Com contacting at the subject's occipital bone, using only the neck and upper back muscles (Schüldt, 1988). This cervical spine extension was useful in generating a maximum exertion from the cervical erector spinae. The next contractions involved lateral flexion of the head to the right and left sides with resistance at the level of the temples (Schüldt, 1988). Forward pushing of the head, with resistance against the forehead, generated maximum activity from the sternocleidomastoid (Schüldt, 1988). In order to generate maximum contractions from the trapezius muscles two maximum exertions were performed. In the first trial the subjects attempted to abduct their arm against resistance while their arm was in a 90° abducted posture (Hägg *et al*, 1987). The second method, shoulder elevation, was achieved by having the subjects pull up on two inextensible straps that were secured to a platform on which the subject stood (Glasscock *et al*, in review).

After the maximum exertions, the subject was taken to the experimental VDT workstation. The monitor was placed at either 15° or 40° below the horizontal plane of the eye. The subject was positioned in the workstation and asked to perform a 2 h reading comprehension task. A script of basic instructions was read to each subject, describing in detail how to perform the task and instructing the subject to refrain from leaning on the workstation but to move head and neck freely as desired. Although no explicit instructions concerning trunk posture were provided, subjects were observed to remain resting against the backrest, which was fixed at 95° (slightly tilted back). Performance was measured by grading the answers given by subjects at the end of each chapter. Subjects were unable to return to any previously read section; therefore, percentage of correct answers provided a reliable task comparison between the two sessions. Pace was also calculated on all subjects by recording the time required to read each chapter and to complete each question and answer section.

At 5 min intervals during the reading task muscle EMG was sampled for 1 s at a frequency of 1024 Hz. If the operator happened to be using the mouse or the keyboard at the time of data collection, the experimenter waited until he or she had completed that activity before collecting EMG data. All EMG data from the 10 muscles were visually inspected on-line after each collection to assure data quality.

Submaximal test contractions were performed every 15 min in order to detect the onset of fatigue in the muscles. The test contraction consisted of a 1 s, non-fatiguing static exertion that interrupted the experimental task for approximately 30 s. The test was conducted with both upper arms abducted 90°, the forearms horizontal and 90° of flexion in the elbows (Hägg *et al*, 1987). A weight equaling 20% of the subject's maximum voluntary force was suspended from each arm, by a sling located just proximal to each elbow. At the same time, the subject exerted 20% of his or her maximum voluntary head extension force with resistance from the Kin Com at the occiput. When the subject reached steady state (usually within 3 s), with the head extension force at 20% and both arms in the correct position, EMG at a frequency of 1024 Hz was collected for 1 s. The weights were then removed from the arms and the subject immediately returned to the task.

A video tape of the subject was used to record postural changes of the subject throughout the session. From this tape, head tilt was sampled at three different times in the experiment (5, 65, 110 min). Each sample was recorded at three time points, 30 s before the sample time, at the sample time and 30 s after the sample time. All time points were then averaged to obtain the mean ear–eye line angle for each subject at each condition. The number of posture shifts, horizontal movements, and the blink rate were also measured from the video tapes and sampled for a period of 5 min at three times (5–10, 65–70, 110–115 min) into the experiment.

At the completion of the 2 h reading task, the subject's visual acuity was immediately tested using the Landholt rings test. The battery of seven maximum voluntary contractions were repeated. Finally, the subjects were asked to complete the subjective assessment/discomfort questionnaire.

Session #2. Session #2 was performed with the monitor in the alternate position. The subject was again provided with the consent form for review and signature. All of the activities performed in Session #1 were repeated with the exception of taking anthropometric measurements. Upon conclusion the subject was asked to complete Part II of the questionnaire. This questionnaire was formatted identically to Part I but with the addition of a question asking which of the monitor positions the subject preferred.

Data analysis

C programs were developed and utilized to process each subject's integrated EMG data and calculate normalized EMG signals. Fast Fourier transform programs written in MatLab software were used to evaluate raw EMG data. MANOVA and ANOVA techniques compared muscle activation and fatigue levels as a function of monitor height.

The video tapes of each session for each subject were evaluated using video analysis. From this analysis blink rate, postural shifts (anterior–posterior and lateral) and sagittal head tilt angle were acquired. Subjective assessment/discomfort, performance rating variables, visual acuity and heart rate were all taken directly from the associated test. All dependent variables were statistically analysed and compared as a function of monitor height. Analyses of variance were used to detect significant effects of the independent variable and the interactions with each of the dependent variables. The Ryan–Einot–Gabriel–Welsch Multiple *F* Test (REGWF) was used to identify the effects that were found to be significant.

Results

Normalized EMG variables

The mean normalized EMG values for all muscles sampled from all subjects during both experimental conditions are summarized in *Table 1*. Analysis of variance results showed that the monitor placement condition had a significant effect ($p < 0.01$) on several of the muscles. A REGWF Test was performed on the data after the ANOVA in order to group the means as equal or significantly different. The 40° condition showed significantly higher mean muscle activity for RSTERNO, RLEV, RCERV, LCERV, RTHOR, and LTHOR. *Figure 4* shows the response of the cervical and thoracic erector spinae muscles. In the 15° condition, only the LTRAP was significantly higher. No significant difference occurred between the 15° and 40° condition for the LSTERNO, LLEV or RTRAP.

Correlation analysis was performed on the data, and no two muscles demonstrated a high correlation coefficient. This demonstrates sufficient isolation of electrode sampling of EMG data.

Muscle fatigue

Raw electromyography data were analysed in the frequency range of 1–512 Hz using a FFT algorithm performed on 1024 data points. These data points were collected during the test contractions over a period of 1 s. The 59–61 Hz frequency bands were removed from the frequency range before analysis. Median frequencies were generated for each muscle, subject and test contraction. Median frequency shift analysis was performed by averaging the initial two-test contraction median values and the final two-test contraction median values. ANOVA

Table 1 Mean values from normalized EMG results

Muscle	REGWF Group	Mean EMG (% of MVC)	Condition	Significance ($p<$)
RSTERNO	A	3.9	40	0.0001
	B	2.7	15	
LSTERNO	A	3.0	15	0.5844
	A	2.9	40	
RLEV	A	2.9	40	0.0001
	B	2.4	15	
LLEV	A	3.5	40	0.4473
	A	3.4	15	
RTRAP	A	2.1	40	0.1716
	A	1.9	15	
LTRAP	A	2.5	15	0.0001
	B	1.9	40	
RCERV	A	6.0	40	0.0001
	B	3.5	15	
LCERV	A	5.8	40	0.0001
	B	3.6	15	
RTHOR	A	5.0	40	0.0001
	B	3.4	15	
LTHOR	A	6.8	40	0.0001
	B	3.3	15	

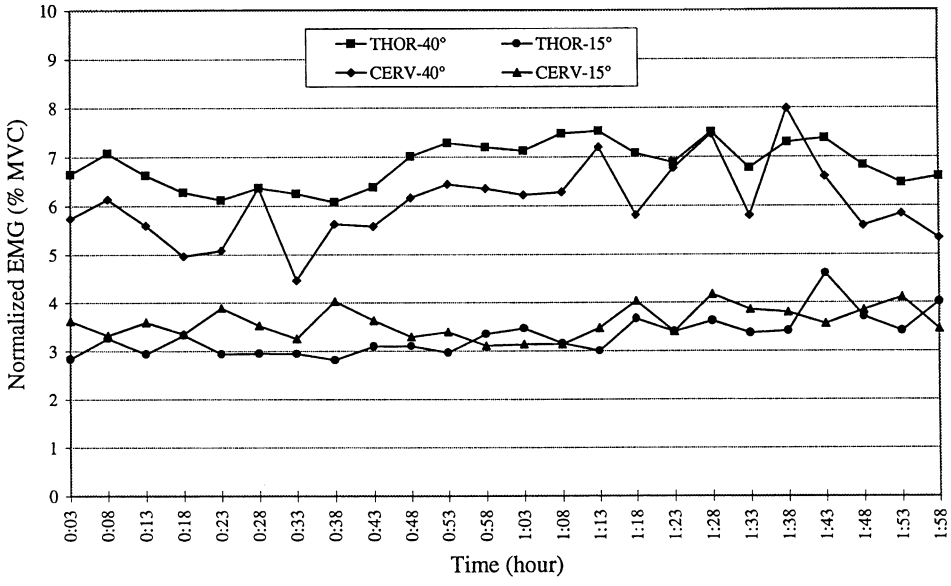


Figure 4 Normalized EMG for cervical and thoracic erector spinae muscles

results showed no significant shift in median frequency over the course of the testing session and no significant interaction of time and monitor condition. No significant fatigue measures were found for either monitor position using this methodology.

Heart rate

The mean heart rate values at each collection period were compared between the two monitor placement conditions. No significant difference was noted between the 40° and 15° placement. Further, no trends over time were apparent in the data, disclosing that heart rate was not affected by monitor position or length of time on task.

Posture variables

At the 15° monitor condition the average head tilt was found to be 3.1° (SD = 5.76) below horizontal. This head tilt was significantly different ($p < 0.0001$) from the 21.3° (SD = 7.93) head tilt obtained at the 40° condition. Thus on average, subjects tilted their heads more than 18° further at the 40° condition than the 15° condition.

The number of posture shifts, also measured from the video, did not significantly differ across the two conditions at any of the three times, nor did they differ between conditions when averaged across time. However, posture shifts did significantly increase over time ($p < 0.05$) for each condition. Further analysis showed that posture shifts in Time 1 differed from posture shifts in Time 3. Posture shifts at Time 2 did not significantly differ from the other two times, though the number of movements fell between those at Time 1 and Time 3. Average number of movements at the three times are displayed in *Table 2*.

Blink rate

Blink rate (blinks/min) was analysed for nine subjects at three different times during both sessions. The rates were then compared across monitor heights and over time. No significant differences were found between the two monitor conditions. Results also did not differ significantly over time. Use of contact lenses was also taken into account; however, no significant differences existed between blink rates of contact wearers and non-wearers.

Table 2 Mean number of postural shifts over time for each condition

	Time 1		Time 2		Time 3	
Condition	**Mean**	**SD**	**Mean**	**SD**	**Mean**	**SD**
15°	2.9	3.2	3.7	3.0	5.8	2.8
40°	3.4	3.4	4.4	2.1	4.0	2.9

Visual acuity

Visual acuity scores were measured on nine subjects following each session. No significant difference was found between scores at the two monitor heights. There was also no significant difference in scores of four contact lens wearers with the five non-wearers. None of the subjects wore glasses.

Performance variables

Performance indices of reading performance and pace for 10 subjects were evaluated for both the 15 and 40° monitor heights. No significant difference in either performance measure was found to exist between the two monitor heights, suggesting that monitor height had no effect on subjects' ability or ease in reading and understanding the material.

Subjective comments and opinions

Seven of the 12 subjects preferred the 15° position. Two of these subjects preferred it because they were more accustomed to the monitor in this position, while two subjects felt the 15° position was more comfortable. One subject cited less visual fatigue at the 15° position and the remaining two subjects did not state the reason why they preferred the 15° position. Of the five subjects that preferred the 40° position, two reported that it allowed their

Table 3 Summary of subjective complaints

Body part	Condition	# Complaints	Average complaint rating
Lower back	15	6	2.83
Lower back	40	6	2.17
Neck	15	5	2.00
Neck	40	5	3.00
Shoulder	15	5	3.00
Shoulder	40	5	2.20
Eye	15	5	1.80
Eye	40	3	2.00
Hips/thighs	15	4	3.25
Hips/thighs	40	3	3.00
Upper back	15	1	3.00
Upper back	40	3	2.67
Knees/feet	15	3	1.67
Knees/feet	40	2	2.50
Wrist/arm	15	2	3.00
Wrist/arm	40	0	0.00
Head	15	1	1.00
Head	40	1	1.00

necks to be more relaxed, two felt it was easier to read and the other subject preferred the 40° position because she felt it was further away, though viewing distance was the same for both monitor positions. Responses to the open-ended questions concerning preference and discomfort revealed eight subjects in each condition reporting discomfort and muscular fatigue in the upper legs and buttocks from sitting for 2 h.

A total of 60 subjective complaints of pain and discomfort were reported by subjects on the discomfort survey ranging from 1 (barely noticeable) to 5 (painful). The lower back received six complaints for each condition and was the most commonly reported body part on the survey. Discomfort in the hips and thighs received the highest rating among subjects with an average complaint rating of 3.25 in the 15° condition and 3.00 in the 40° condition. Complaint frequency, average rating per complaint, and body part/condition are displayed in *Table 3*. No significance was found for any subjective variable as a function of monitor position, yet these results reflect the large number of complaints usually encountered when investigating VDT stations.

Discussion

The results of the normalized EMG analysis showed significantly higher muscle activity in the 40° monitor condition for the cervical and thoracic erector spinae. The magnitude of EMG activity doubled from 3% maximum voluntary contraction (MVC) in the 15° condition to 6% MVC in the 40° condition (*Figure 4*). According to Jonsson (1978) static load level should not exceed 5% MVC for work tasks of long duration. Additionally, Sjøgaard *et al* (1986) found that static contractions of 5% MVC sustained for 1 h induced muscle fatigue as indicated by intramuscular pressure. Although subjects were allowed to move throughout the experiment, operating a VDT is defined as a static task. The FFT analysis of muscle fatigue did not show onset of fatigue for either condition. This may be due to the relatively short duration of the sessions, 2 h, versus a typical 8 h work day.

The left trapezius was the only muscle that showed higher activity at the 15° condition. The magnitude of EMG activity for both conditions was substantially less than the 5% MVC level shown to induce fatigue. The 15° condition exhibited a 2.5% MVC while the 40° condition was at 1.9%. This activity could be a result of the posture assumed between the two conditions. At the 15° condition, most subjects were sitting more erect, while at the 40° position, the shoulders tended to relax forward. The right trapezius muscle activity levels were comparable, possibly resulting from the use of the right arm for scrolling through the text in both conditions.

Analysis of the video tapes for each session revealed a significant increase ($p < 0.05$) over time in all body movements for both conditions. Between the two monitor placement conditions however, no significant difference was found in the total number of movements throughout the experiment. These results contradict Ankrum and Nemeth (1995) who suggested that a lower monitor placement would encourage movement as users could assume a wider variety of postures without sacrificing visual acuity. Instead, our results reveal the increased desire to shift postures in both conditions over the duration of the task, thus reflecting the discomfort reported in subjective questioning.

Video analysis further allowed for quantification of head tilt. The average head tilt, measured below the horizontal, at the 40° condition was -21.3 (SD = 8.7)° and 3.1 (SD = 6.8)° during the 15° condition. The 40° condition was found to necessitate a significantly greater head tilt than the 15° condition. A lower monitor height is recommended for its flexibility in allowing the user to assume either a lower gaze angle or greater head tilt (Ankrum and Nemeth, 1995). Our findings indicate that subjects at the lower monitor condition are not shifting between the postures that are theoretically available and are instead assuming a relatively static deviated posture. In a previous study, Chaffin (1973) found no musculoskeletal problems when the head tilt was less than 15°, but when the head tilt increased to 30°, fatigue was apparent. However, there has been insufficient evidence of fatigue within the range of 15° and 30° of head tilt. The 21.3° of average head tilt found at the 40° condition falls into this range. Although fatigue was not present in the FFT analysis, the significantly elevated muscle activity in the erector spinae indicates that the muscles were working harder to maintain the deviated head position. Future work that includes a more realistic 8 h work day may be necessary to elicit the longer term effects of muscular fatigue, visual fatigue, degraded operator performance and distinct changes in operator physiology.

Conclusions

As the use of VDTs increases in the home, school and workplace, the number of related physical discomfort complaints is likely to increase. Many have suggested alternate workstation configurations in an attempt to alleviate physical factors promoting discomfort. One such suggestion is lowering the height of the video display terminal, which has been argued to reduce visual fatigue, allow the user to assume a wider variety of postures and reduce static loading. In this study we compared a 15° and 40° below horizontal monitor placement and investigated the correlating physical differences during an experimental session. The expressed advantages of the lower monitor were not found in this 2 h

experiment. Subjects did not shift their postures more with either monitor condition nor was blink rate or visual acuity found to be significantly different between the conditions. Greater muscle activity for six of the ten muscles investigated was found during the 40° position. Although corresponding muscular fatigue was not detected, the activity of six muscles of the neck and back was found to double for the 40° condition compared to the 15° condition. Therefore, the results of this 2 h reading comprehension study found no evidence in any of the dependent measures to indicate that switching from a 15° below horizontal monitor placement to a 40° monitor position would improve the health and safety of the operator.

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